Kyra Movva Student at American Heritage

Github: https://github.com/Kyra15
Email: kyra.movva@gmail.com
Linkedin: https://www.linkedin.com/in/kyramovva/
Itch: https://kyramovva15.itch.io/

SKILLS

- **Programming languages:** Python (proficient), GDScript, Java, HTML/CSS (intermediate), Javascript, C, SQL (beginner); **Libraries**: Flask API, Numpy/Pandas, Pytorch, Tensorflow, Streamlit, Tkinter, OpenCV
- **Concepts**: APIs, Regex commands, GUIs, Image Processing, Basic Networking, Graphic Design, Bluetooth Low Energy
- **Platforms**: VSCode, PyCharms, Godot, Arduino Uno; **Hardware**: RaspberryPi, DC Motor Drivers, Arduino, Breadboarding

EXPERIENCE

Software Intern at CarePredict

June 2024

- Interned for 1 month at a local computer science and health care start-up
- Created app for data graphing and visualization
- Worked with data management

Elementary Science Olympiad Coach

September 2022 - Present

- Coached elementary school students to top 3 finishes in state competition in subjects such as programming, cryptography, and astronomy

Computer Science Tutor for the Computer Science Society

September 2023 - Present

- Tutored students in creating algorithms, coding logic, troubleshooting and bug fixes

CoderGals Instructor

February 2025 - Present

- Teach elementary school girls Python for 2 hours every week
- Create slides and problems for the girls to do
- Appointed Vice-President (11th grade)
- Devised curriculum plan for students to create video games as final projects

PROJECTS

Self-Driving Tank with GUI

- Done during Programming with a Purpose at American Heritage
- The tank API uses HTTP commands and is accessed by a Raspberry Pi via TCP/IP, with obstacle detection, image recognition, manual controls, and a live webcam video feed with a live overlay of the lane markers

- Uses SQLite database to store user data, Flask for the API, OpenCV for the image recognition, and Tkinter for the GUI
- Primarily uses a Python GUI client, but also coded a Java GUI client and HTML GUI

Paddlemania!

- Endless runner video game built within a few weeks using the Godot game engine for the 2024 Computer Science Fair at American Heritage
- Featured a player, enemies, and an in-game currency and timer system for points as well as full collision detection for all entities and multiple pages including shopping and settings pages with full external game state save.

Poverello Food Bank Inventory

- Programmed a HTML, CSS, Javascript website for a non-profit food bank
- Was the project manager for the front end team of the inventory website
- Utilized a Flask API and SQLite for data management

Cursive Handwriting Analysis

- Done during Machine Learning and AI at American Heritage
- Developed CNN to identify handwritten cursive digits from real world data
- Utilized OpenCV and PIL for data processing and Pytorch to build the CNN
- Achieved a test accuracy of roughly 90%

Tetris AI Agent

- Done during Machine Learning and AI at American Heritage
- Programmed a Genetic AI agent in Python for playing tetris
- Achieved a highest time played of 49.7 hours

On the Search for a New Earth

- Done during UC Berkeley Astro 9
- Created a website to check the habitability of exoplanets via lightcurve modelling using astropy and lightkurve

American Heritage Annual Game Jam

- Created and planned a themed game jam event at my school
- Ran successfully for 2 years so far, accumulating a total participation count of over 60+ with 9 total games

COURSES

GPA: 6.0

- Computer Science Essentials
- Programing with a Purpose
- AP Computer Science A
- AP Statistics
- AP Physics 1
- AP Physics C: Mechanics
- AP Calculus AB
- UC Berkeley: Python in Astronomy
- Harvard: AI Bootcamp
- NYU: Machine Learning

- AP Precalculus
- Machine Learning/Cybersecurity

LEADERSHIP

Computer Science Society:

- Co-Secretary (10th), Co-President (11th, 12th)
- Managed team of 20+ officers and 130 members, educating students on computer science and fundraising $1k+

Astronomy Club:

- Vice-President (10th), Co-President (11th, 12th)
- Educated students on astronomy weekly
- Currently co-founding school rocketry team to compete in the American Rocketry Challenge

Science Olympiad:

- Secretary (11th)
- Competed in various physical science related events
- 1st place finishes at national tournaments in cryptography, geology, astronomy, and physics

COMMUNITY SERVICE

Food Bank Volunteering:

- Volunteer at a Food Bank + Donations - 35 hours

Science Olympiad Coach:

- Science instructor and chaperone for elementary - 300+ hours

Codergals Teacher:

- Taught elementary girls computer science - 40+ hours

AWARDS

International Astronomy and Astrophysics Competition - 2025

- Awarded gold honor after 3 rounds of rigorous astrophysics and research
- Top 2% globally in astronomy/astrophysics (1 of 14 US recipients)

Science Olympiad - 2023 - Present

- Florida Regionals:
  - 1st place - Experimental Design, Codebusters, Rocks and Minerals, Machines
- Florida States:
  - 1st place - Codebusters, Rocks and Minerals
- Caltech National Invitational:
  - 3rd place - Rocks and Minerals
- Harvard National Invitational:
  - 1st place - Codebusters
- Carnegie Mellon University National Invitational:
  - 3rd place - Codebusters

Competitive Programming - 2025 - Present

- 5th place out of 76 teams at UCF National Hackathon (data structures and algorithms)
- USA Coding Olympiad Silver Level
- 2nd @ American Heritage Hackathon (engineering, computer science, and physics)

Math Competition - 2022 - Present

- American Math Competition (AMC) score: 88.5 on AMC 12B 2025
- 6th place in the Florida geometry spring season (squirty chicken award)
- Top 10 finishes at state and national conventions in Geometry
- 2nd place @ national convention - Gemini
- 1st place @ state convention - Math Poster
- 1st place @ state convention - Digital Scrapbook
- 6th place @ state convention - Computer Competition